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ACME PETROCHEMICAL DEMO COMPLEX - UNIT 2300

PIPING & INSTRUMENTATION DIAGRAM: SIS CAUSE & EFFECT INTERLOCK MATRIX

Drawing Number: PID-23-0002 | Revision: Z1 | Safety Integrity Level: SIL-2 (IEC 61511)

1. INTERLOCK I-2304: DECOMPOSER REACTOR D-2304 EXOTHERMIC RUNAWAY PROTECTION (SIL-2)

Initiating Causes (Any of the following):

- Cause 1A: D-2304 Reactor Temperature High-High TXSHH-1301 ≥ 115.0 degC (2oo3 voting TXSHH-1301A/B/C)
- Cause 1B: D-2304 Reactor Pressure High-High PAHH-1301 ≥ 5.5 kg/cm²g (PT-1301)
- Cause 1C: D-2304 Recirculation Flow Low-Low FSLL-1304 $\leq 110,000$ kg/hr (FT-1304)

Automated Trip Effects (Executed within ≤ 2.0 seconds):

- Effect 1: Trip close Sulfuric Acid Catalyst Isolation Valve XV-1305 (stops acid catalyst feed Stream S310)
- Effect 2: Trip open Emergency Cumene Quench Valve XV-1309 (injects cold cumene Stream S387 into D-2304 to dilute)
- Effect 3: Trip stop Preflash Column Bottoms Pump P-2301A/B (stops concentrated 82 wt% CHP feed Stream S229)
- Effect 4: Force E-2307 Cooling Water Control Valve TV-1302 to 100% Open (FO)

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PID-23-0002 PAGE 2 OF 3 - INTERLOCK I-2301: PREFLASH COLUMN V-2301 THERMAL STABILITY TRIP (SIL-2)

Initiating Causes:

- Cause 2A: V-2301 Sump Temperature High-High TAAH-0402 $\geq$ 105.0 degC (1oo2 voting)
- Cause 2B: V-2301 Overhead Vacuum Loss PAHH-0401 $\geq$ 75.0 mmHgA (PT-0401 on X-2301 suction)
- Cause 2C: V-2301 Sump Level High-High LSHH-0401 $\geq$ 78.0% (LT-0401)

Automated Trip Effects:

- Effect 1: Trip close Reboiler Heating Steam Isolation Valve XV-0403
- Effect 2: Open Emergency Dilute Cumene Flush Valve XV-0409 into V-2301 Sump (Nozzle N6)

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PID-23-0002 PAGE 3 OF 3 - SIS PROOF TEST & BYPASS GOVERNANCE

- Proof Test Interval: 12 Months (Aligned with STD-PHA-001 LOPA Credit $PFD_{avg} \leq 5.0e-3$ for SIL-2)
- Maintenance Bypass Alarm: Any Force/Bypass on TXSHH-1301 or XV-1309 triggers continuous DCS Priority-1 announcement